

CHAPTER 1

THE COMPELLING NEED FOR DATA WAREHOUSING

CHAPTER OBJECTIVES

- Understand the desperate need for strategic information
- Recognize the information crisis at every enterprise
- Distinguish between operational and informational systems
- Learn why all past attempts to provide strategic information failed
- Clearly see why data warehousing is the viable solution
- Understand business intelligence for an enterprise

As an information technology (IT) professional, you have worked on computer applications as an analyst, programmer, designer, developer, database administrator, or project manager. You have been involved in the design, implementation, and maintenance of systems that support day-to-day business operations. Depending on the industries you have worked in, you must have been involved in applications such as order processing, general ledger, inventory, human resources, payroll, in-patient billing, checking accounts, insurance claims, and so on.

These applications are important systems that run businesses. They process orders, maintain inventory, keep the accounting books, service the clients, receive payments, and process claims. Without these computer systems, no modern business can survive. Companies started building and using these systems in the 1960s and have become completely dependent on them. As an enterprise grows larger, hundreds of computer applications are needed to support the various business processes. These applications are effective in what they are designed to do. They gather, store, and process all the data needed to successfully perform the daily routine operations. They provide online information and produce a variety of reports to monitor and run the business.

Organizations achieve competitive advantage:

◆ Retail	◆ Manufacturing
◆ Customer Loyalty	◆ Cost Reduction
◆ Market Planning	◆ Logistics Management
◆ Financial	◆ Utilities
◆ Risk Management	◆ Asset Management
◆ Fraud Detection	◆ Resource Management
◆ Airlines	◆ Government
◆ Route Profitability	◆ Manpower Planning
◆ Yield Management	◆ Cost Control

Figure 1-1 Organizations' use of data warehousing.

In the 1990s, as businesses grew more complex, corporations spread globally, and competition became fiercer, business executives became desperate for information to stay competitive and improve the bottom line. The operational computer systems did provide information to run the day-to-day operations but what the executives needed were different kinds of information that could be used readily to make strategic decisions. The decision makers wanted to know which geographic regions to focus on, which product lines to expand, and which markets to strengthen. They needed the type of information with proper content and format that could help them make such strategic decisions. We may call this type of information strategic information as different from operational information. The operational systems, important as they were, could not provide strategic information. Businesses, therefore, were compelled to turn to new ways of getting strategic information.

Data warehousing is a new paradigm specifically intended to provide vital strategic information. In the 1990s, organizations began to achieve competitive advantage by building data warehouse systems. Figure 1-1 shows a sample of strategic areas where data warehousing had already produced results in different industries.

At the outset, let us now examine the crucial question: why do enterprises really need data warehouses? This discussion is important because unless we grasp the significance of this critical need, our study of data warehousing will lack motivation. So, please pay close attention.

ESCALATING NEED FOR STRATEGIC INFORMATION

While we discuss the clamor by enterprises for strategic information, we need to look at the prevailing information crisis that was holding them back, as well as the technology trends of the past few years that are working in our favor, enabling us to provide strategic information. Our discussion of the need for strategic information will not be complete unless we study the opportunities provided by strategic information and the risks facing a company without such information.

Who needs strategic information in an enterprise? What exactly do we mean by strategic information? The executives and managers who are responsible for keeping the enterprise

competitive need information to make proper decisions. They need information to formulate the business strategies, establish goals, set objectives, and monitor results.

Here are some examples of business objectives:

- Retain the present customer base
- Increase the customer base by 15% over the next 5 years
- Improve product quality levels in the top five product groups
- Gain market share by 10% in the next 3 years
- Enhance customer service level in shipments
- Bring three new products to market in 2 years
- Increase sales by 15% in the North East Division

For making decisions about these objectives, executives and managers need information for the following purposes: to get in-depth knowledge of their company’s operations, review and monitor key performance indicators and note how these affect one another, keep track of how business factors change over time, and compare their company’s performance relative to the competition and to industry benchmarks. Executives and managers need to focus their attention on customers’ needs and preferences, emerging technologies, sales and marketing results, and quality levels of products and services. The types of information needed to make decisions in the formulation and execution of business strategies and objectives are broad-based and encompass the entire organization. All these types of essential information may be combined under the broad classification called strategic information.

Strategic information is not for running the day-to-day operations of the business. It is not intended to produce an invoice, make a shipment, settle a claim, or post a withdrawal from a bank account. Strategic information is far more important for the continued health and survival of the corporation. Critical business decisions depend on the availability of proper strategic information in an enterprise. Figure 1-2 lists the desired characteristics of strategic information.

INTEGRATED	Must have a single, enterprise-wide view.
DATA INTEGRITY	Information must be accurate and must conform to business rules.
ACCESSIBLE	Easily accessible with intuitive access paths, and responsive for analysis.
CREDIBLE	Every business factor must have one and only one value.
TIMELY	Information must be available within the stipulated time frame.

Figure 1-2 Characteristics of strategic information.

The Information Crisis

You may be working in the IT department of a large conglomerate or you may be part of a medium-sized company. Whatever may be the size of your company, think of all the various computer applications in your company. Think of all the databases and the quantities of data that support the operations of your company. How many years' worth of customer data is saved and available? How many years' worth of financial data is kept in storage? Ten years? Fifteen years? Where is all this data? On one platform? In legacy systems? In client/server applications?

We are faced with two startling facts: (1) organizations have lots of data, (2) information technology resources and systems are not effective at turning all that data into useful strategic information. Over the past two decades, companies have accumulated tons and tons of data about their operations. Mountains of data exist. Information is said to double every 18 months.

If we have such huge quantities of data in our organizations, why can't our executives and managers use this data for making strategic decisions? Lots and lots of information exists. Why then do we talk about an information crisis? Most companies are faced with an information crisis not because of lack of sufficient data, but because the available data is not readily usable for strategic decision making. These large quantities of data are very useful and good for running the business operations but hardly amenable for use in making decisions about business strategies and objectives.

Why is this so? First, the data of an enterprise is spread across many types of incompatible structures and systems. Your order processing system might have been developed 25 years ago and is still running on an old mainframe. Possibly, some of the data may still be on VSAM files. Your later credit assignment and verification system might be on a client/server platform and the data for this application might be in relational tables. The data in a corporation resides in various disparate systems, multiple platforms, and diverse structures. The more technology your company has used in the past, the more disparate the data of your company will be. But, for proper decision making on overall corporate strategies and objectives, we need information integrated from all systems.

Data needed for strategic decision making must be in a format suitable for easy analysis to spot trends. Executives and managers need to look at trends over time and steer their companies in the proper direction. The tons of available operational data cannot be readily used to discern trends. Operational data is event-driven. You get snapshots of transactions that happen at specific times. You have data about units of sale of a single product in a specific order on a given date to a certain customer. In the operational systems, you do not readily have the trends of a single product over the period of a month, a quarter, or a year.

For strategic decision making, executives and managers must be able to review data from different business viewpoints. For example, they must be able to review and analyze sales quantities by product, salesperson, district, region, and customer groups. Can you think of operational data being readily available for such analysis? Operational data is not directly suitable for review from different viewpoints.

Technology Trends

Those of us who have worked in the information technology field for two or three decades have witnessed the breathtaking changes that have taken place. First, the name of the computer department in an enterprise went from "data processing" to "management information

systems,” then to “information systems,” and more recently to “information technology.” The entire spectrum of computing has undergone tremendous changes. The computing focus itself has changed over the years. Old practices could not meet new needs. Screens and preformatted reports are no longer adequate to meet user requirements.

Over the years, the price of MIPS (million instructions per second) is continuing to decline, digital storage is costing less and less, and network bandwidth is increasing as its price decreases. Specifically, we have seen explosive changes in these critical areas:

- Computing technology
- Human–machine interface
- Processing options

Figure 1-3 illustrates these waves of explosive growth.

What is our current position in the technology revolution? Hardware economics and miniaturization allow a workstation on every desk and provide increasing power at reducing costs. New software provides easy-to-use systems. Open systems architecture creates cooperation and enables the use of multivendor software. Improved connectivity, networking, and the Internet open up interaction with an enormous number of systems and databases.

All of these improvements in technology are meritorious. These have made computing faster, cheaper, and widely available. But what is their relevance to the escalating need for strategic information? Let us understand how the current state of the technology is conducive to providing strategic information.

Providing strategic information requires collection of large volumes of corporate data and storing it in suitable formats. Technology advances in data storage and reduction in storage costs readily accommodate data storage needs for strategic decision-support systems. Analysts, executives, and managers use strategic information interactively to analyze and spot business trends. The user will ask a question and get the results, then ask another question, look at the results, and ask yet another question. This interactive process continues. Tremendous advances in interface software make such interactive analysis possible.

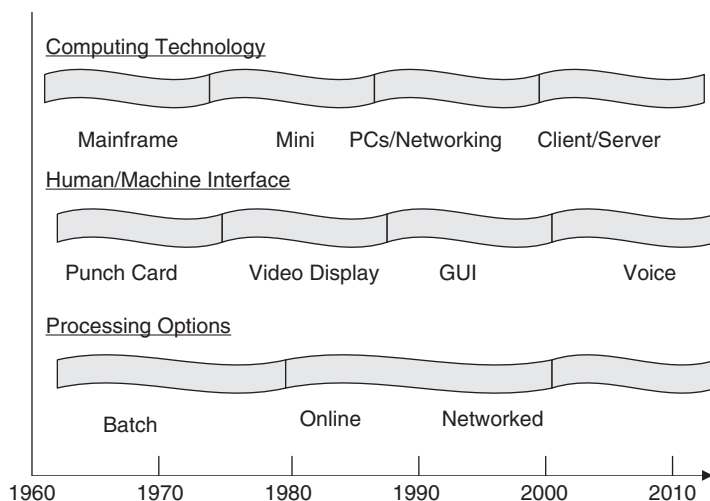


Figure 1-3 Explosive growth of information technology.

Processing large volumes of data and providing interactive analysis requires extra computing power. The explosive increase in computing power and its lower costs make provision of strategic information feasible. What we could not accomplish a few years earlier for providing strategic information is now possible with the current advanced stage of information technology.

Opportunities and Risks

We have looked at the information crisis that exists in every enterprise and grasped that in spite of lots of operational data in the enterprise, data suitable for strategic decision making is not available. Yet, the current state of the technology can make it possible to provide strategic information. While we are still discussing the escalating need for strategic information by companies, let us ask some basic questions. What are the opportunities available to companies resulting from the possible use of strategic information? What are the threats and risks resulting from the lack of strategic information available in companies?

Here are some examples of the opportunities made available to companies through the use of strategic information:

- A business unit of a leading long-distance telephone carrier empowers its sales personnel to make better business decisions and thereby capture more business in a highly competitive, multibillion-dollar market. A Web-accessible solution gathers internal and external data to provide strategic information.
- Availability of strategic information at one of the largest banks in the United States with assets in the \$250 billion range allows users to make quick decisions to retain their valued customers.
- In the case of a large health management organization, significant improvements in health care programs are realized, resulting in a 22% decrease in emergency room visits, 29% decrease in hospital admissions for asthmatic children, potentially sight-saving screenings for hundreds of diabetics, improved vaccination rates, and more than 100,000 performance reports created annually for physicians and pharmacists.
- At one of the top five U.S. retailers, strategic information combined with Web-enabled analysis tools enables merchants to gain insights into their customer base, manage inventories more tightly, and keep the right products in front of the right people at the right place at the right time.
- A community-based pharmacy that competes on a national scale with more than 800 franchised pharmacies coast to coast gains in-depth understanding of what customers buy, resulting in reduced inventory levels, improved effectiveness of promotions and marketing campaigns, and improved profitability for the company.
- A large electronics company saves millions of dollars a year because of better management of inventory.

On the other hand, consider the following cases where risks and threats of failures existed before strategic information was made available for analysis and decision making:

- With an average fleet of about 150,000 vehicles, a nationwide car rental company can easily get into the red at the bottom line if fleet management is not effective. The fleet is the biggest cost in that business. With intensified competition, the potential for failure is immense if the fleet is not managed effectively. Car idle time must be kept to an

absolute minimum. In attempting to accomplish this, failure to have the right class of car available in the right place at the right time, all washed and ready, can lead to serious loss of business.

- For a world-leading supplier of systems and components to automobile and light truck equipment manufacturers, serious challenges faced included inconsistent data computations across nearly 100 plants, inability to benchmark quality metrics, and time-consuming manual collection of data. Reports needed to support decision making took weeks. It was never easy to get company-wide integrated information.
- For a large utility company that provided electricity to about 25 million consumers in five mid-Atlantic states in the United States, deregulation could result in a few winners and lots of losers. Remaining competitive and perhaps even just surviving depended on centralizing strategic information from various sources, streamlining data access, and facilitating analysis of the information by the business units.

FAILURES OF PAST DECISION-SUPPORT SYSTEMS

Assume a specific scenario. The marketing department in your company has been concerned about the performance of the West Coast region and the sales numbers from the monthly report this month are drastically low. The marketing vice president is agitated and wants to get some reports from the IT department to analyze the performance over the past two years, product by product, and compared to monthly targets. He wants to make quick strategic decisions to rectify the situation. The CIO wants your boss to deliver the reports as soon as possible. Your boss runs to you and asks you to stop everything and work on the reports. There are no regular reports from any system to give the marketing department what they want. You have to gather the data from multiple applications and start from scratch. Does this sound familiar?

At one time or another in your career in information technology, you must have been exposed to situations like this. Sometimes, you may be able to get the information required for such ad hoc reports from the databases or files of one application. Usually this is not so. You may have to go to several applications, perhaps running on different platforms in your company environment, to get the information. What happens next? The marketing department likes the ad hoc reports you have produced. But now they would like reports in a different format, containing more information that they did not think of originally. After the second round, they find that the contents of the reports are still not exactly what they wanted. They may also find inconsistencies among the data obtained from different applications.

The fact is that for nearly two decades or more, IT departments have been attempting to provide information to key personnel in their companies for making strategic decisions. Sometimes an IT department could produce ad hoc reports from a single application. In most cases, the reports would need data from multiple systems, requiring the writing of extract programs to create intermediary files that could be used to produce the ad hoc reports.

Most of these attempts by IT in the past ended in failure. The users could not clearly define what they wanted in the first place. Once they saw the first set of reports, they wanted more data in different formats. The chain continued. This was mainly because of the very nature of the process of making strategic decisions. Information needed for strategic decision making has to be available in an interactive manner. The user must be able to query online, get results, and query some more. The information must be in a format suitable for analysis.

In order to appreciate the reasons for the failure of IT to provide strategic information in the past, we need to consider how IT was attempting to do this all these years. Let us, therefore, quickly run through a brief history of decision support systems.

History of Decision-Support Systems

Depending on the size and nature of the business, most companies have gone through the following stages of attempts to provide strategic information for decision making.

Ad hoc Reports. This was the earliest stage. Users, especially from marketing and finance, would send requests to IT for special reports. IT would write special programs, typically one for each request, and produce the ad hoc reports.

Special Extract Programs. This stage was an attempt by IT to anticipate somewhat the types of reports that would be requested from time to time. IT would write a suite of programs and run the programs periodically to extract data from the various applications. IT would create and keep the extract files to fulfill any requests for special reports. For any reports that could not be run off the extracted files, IT would write individual special programs.

Small Applications. In this stage, IT formalized the extract process. IT would create simple applications based on the extracted files. The users could stipulate the parameters for each special report. The report printing programs would print the information based on user-specific parameters. Some advanced applications would also allow users to view information through online screens.

Information Centers. In the early 1970s, some major corporations created what were called information centers. The information center typically was a place where users could go to request ad hoc reports or view special information on screens. These were predetermined reports or screens. IT personnel were present at these information centers to help the users to obtain the desired information.

Decision-Support Systems. In this stage, companies began to build more sophisticated systems intended to provide some semblance of strategic information. Again, similar to the earlier attempts, these systems were supported by extracted files. The systems were menu-driven and provided online information and also the ability to print special reports. Many such decision-support systems were for marketing.

Executive Information Systems. This was an attempt to bring strategic information to the executive desktop. The main criteria were simplicity and ease of use. The system would display key information every day and provide the ability to request simple, straightforward reports. However, only preprogrammed screens and reports were available. After seeing the total countrywide sales, if the executive wanted to see the analysis by region, by product, or by another dimension, it was not possible unless such breakdowns were already preprogrammed. This limitation caused frustration and executive information systems did not last long in many companies.

Inability to Provide Information

Every one of the past attempts at providing strategic information to decision makers was unsatisfactory. Figure 1-4 depicts the inadequate attempts by IT to provide strategic information. As IT professionals, we are all familiar with the situation.

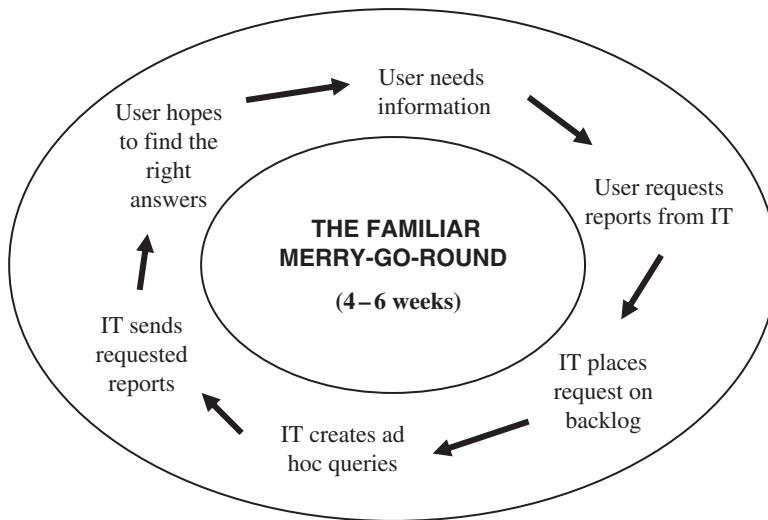


Figure 1-4 Inadequate attempts by IT to provide strategic information.

Here are some of the factors relating to the inability to provide strategic information:

- IT receives too many ad hoc requests, resulting in a large overload. With limited resources, IT is unable to respond to the numerous requests in a timely fashion.
- Requests are too numerous; they also keep changing all the time. The users need more reports to expand and understand the earlier reports.
- The users find that they get into the spiral of asking for more and more supplementary reports, so they sometimes adapt by asking for every possible combination, which only increases the IT load even further.
- The users have to depend on IT to provide the information. They are not able to access the information themselves interactively.
- The information environment ideally suited for strategic decision making has to be very flexible and conducive for analysis. IT has been unable to provide such an environment.

OPERATIONAL VERSUS DECISION-SUPPORT SYSTEMS

Is there an underlying reason for the failure of all the previous attempts by IT to provide strategic information? What has IT been doing all along? The fundamental reason for the inability to provide strategic information is that we have been trying all along to provide strategic information from the operational systems. These operational systems such as order processing, inventory control, claims processing, outpatient billing, and so on are not designed or intended to provide strategic information. If we need the ability to provide strategic information, we must get the information from altogether different types of systems. Only specially designed decision support systems or informational systems can provide strategic information. Let us understand why.

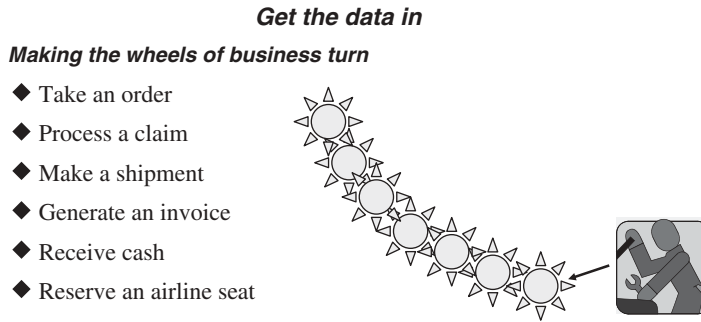


Figure 1-5 Operational systems.

Making the Wheels of Business Turn

Operational systems are online transaction processing (OLTP) systems. These are the systems that are used to run the day-to-day core business of the company. They are the so-called bread-and-butter systems. Operational systems make the wheels of business turn (see Fig. 1-5). They support the basic business processes of the company. These systems typically get the data into the database. Each transaction processes information about a single entity such as a single order, a single invoice, or a single customer.

Watching the Wheels of Business Turn

On the other hand, specially designed and built decision-support systems are not meant to run the core business processes. They are used to watch how the business runs, and then make strategic decisions to improve the business (see Fig. 1-6).

Decision-support systems are developed to get strategic information *out of* the database, as opposed to OLTP systems that are designed to put the data *into* the database. Decision-support systems are developed to provide strategic information.

Different Scope, Different Purposes

Therefore, we find that in order to provide strategic information we need to build informational systems that are different from the operational systems we have been building to run the basic business. It will be worthless to continue to dip into the operational systems for

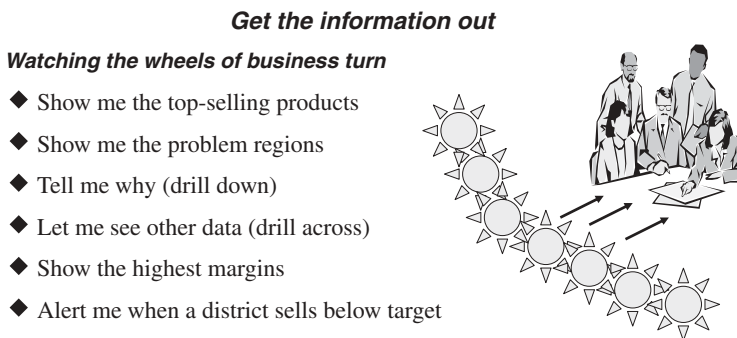


Figure 1-6 Decision-support systems.

How are they different?

	OPERATIONAL	INFORMATIONAL
Data Content	Current values	Archived, derived, summarized
Data Structure	Optimized for transactions	Optimized for complex queries
Access Frequency	High	Medium to low
Access Type	Read, update, delete	Read
Usage	Predictable, repetitive	Ad hoc, random, heuristic
Response Time	Sub-seconds	Several seconds to minutes
Users	Large number	Relatively small number

Figure 1-7 Operational and informational systems.

strategic information as we have been doing in the past. As companies face fiercer competition and businesses become more complex, continuing the past practices will only lead to disaster.

We need to design and build informational systems

- That serve different purposes
- Whose scopes are different
- Whose data content is different
- Where the data usage patterns are different
- Where the data access types are different

Figure 1-7 summarizes the differences between the traditional operational systems and the newer informational systems that need to be built.

DATA WAREHOUSING—THE ONLY VIABLE SOLUTION

At this stage of our discussion, we now realize that we do need different types of decision-support systems to provide strategic information. The type of information needed for strategic decision making is different from that available from operational systems. We need a new type of system environment for the purpose of providing strategic information for analysis, discerning trends, and monitoring performance.

Let us examine the desirable features and processing requirements of this new type of system environment. Let us also consider the advantages of this type of system environment designed for strategic information.

A New Type of System Environment

The desired features of the new type of system environment are:

- Database designed for analytical tasks
- Data from multiple applications
- Easy to use and conducive to long interactive sessions by users

- Read-intensive data usage
- Direct interaction with the system by the users without IT assistance
- Content updated periodically and stable
- Content to include current and historical data
- Ability for users to run queries and get results online
- Ability for users to initiate reports

Processing Requirements in the New Environment

Most of the processing in the new environment for strategic information will have to be analytical. There are at least four levels of analytical processing requirements:

1. Running of simple queries and reports against current and historical data.
2. Ability to perform “what if” analysis in many different ways.
3. Ability to query, step back, analyze, and then continue the process to any desired length.
4. Ability to spot historical trends and apply them in future interactive processes.

Strategic Information from the Data Warehouse

This new system environment that users desperately need to obtain strategic information happens to be the new paradigm of data warehousing. Beginning with the late 1980s and into the 1990s enterprises began building such system environments. This new environment, known as the data warehouse environment, is kept separate from the system environment that supports the routine day-to-day operations. The data warehouse essentially has become the source of strategic information for the enterprise to enable strategic decision making. The data warehouse has proved to be the only viable solution. We have clearly seen that solutions based on the data extracted from operational systems have all been totally unsatisfactory. Figure 1-8 shows the general overview of the data warehouse as the source of strategic information for the enterprise.

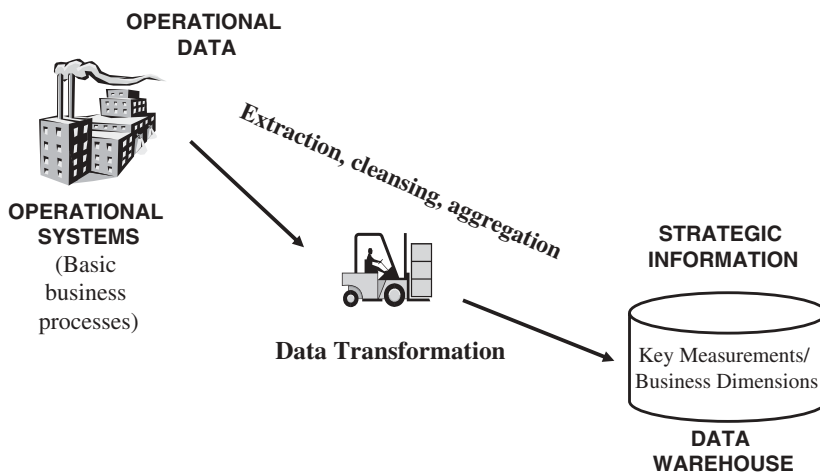


Figure 1-8 General overview of the data warehouse.

At a high level of interpretation, the data warehouse contains critical metrics of the business processes stored along business dimensions. For example, a data warehouse might contain units of sales, by product, day, customer group, sales district, sales region, and promotion. Here the business dimensions are product, day, customer group, sales district, sales region, and sales promotion type. Unit sales represent the metrics being measured across products, days, customer groups, sales districts, sales regions, and sales promotion types.

From where does the data warehouse get its data? The data is derived from the operational systems that support the basic business processes of the organization. In between the operational systems and the data warehouse, there is a data staging area. In this staging area, the operational data is cleansed and transformed into a form suitable for placement in the data warehouse for easy retrieval.

DATA WAREHOUSE DEFINED

We have reached the strong conclusion that data warehousing is the only viable solution for providing strategic information. We arrived at this conclusion based on the functions of the new system environment called the data warehouse. So, let us try to come up with a functional definition of the data warehouse.

The data warehouse is an informational environment that:

- Provides an integrated and total view of the enterprise.
- Makes the enterprise's current and historical information easily available for strategic decision making.
- Makes decision-support transactions possible without hindering operational systems.
- Renders the organization's information consistent.
- Presents a flexible and interactive source of strategic information.

A Simple Concept for Information Delivery

In the final analysis, data warehousing is a simple concept. It is born out of the need for strategic information and is the result of the search for a new way to provide such information. The methods of the previous decades using the operational computing environment were unsatisfactory. The new concept is not to generate fresh data, but to make use of the large volumes of existing data and to transform it into forms suitable for providing strategic information.

The data warehouse exists to answer questions users have about the business, the performance of the various operations, the business trends, and about what can be done to improve the business. The data warehouse exists to provide business users with direct access to data, to provide a single unified version of the key performance indicators, to record the past accurately, and to provide the ability for viewing the data from many different perspectives. In short, the data warehouse is there to support decisional processes.

Data warehousing is really a simple concept: Take all the data you already have in the organization, clean and transform it, and then provide useful strategic information. What could be simpler than that?

An Environment, Not a Product

A data warehouse is not a single software or hardware product you purchase to provide strategic information. It is, rather, a computing environment where users can find strategic

information, an environment where users are put directly in touch with the data they need to make better decisions. It is a user-centric environment.

Let us summarize the characteristics of this new computing environment called the data warehouse:

- An ideal environment for data analysis and decision support
- Fluid, flexible, and interactive
- 100% user-driven
- Very responsive and conducive to the ask–answer–ask again pattern
- Provides the ability to discover answers to complex, unpredictable questions

A Blend of Many Technologies

Let us reexamine the basic concept of data warehousing. The basic concept of data warehousing is:

- Take all the data from the operational systems.
- Where necessary, include relevant data from outside, such as industry benchmark indicators.
- Integrate all the data from the various sources.
- Remove inconsistencies and transform the data.
- Store the data in formats suitable for easy access for decision making.

Different technologies are, therefore, needed to support these functions. Figure 1-9 shows how a data warehouse is a blend of the many technologies needed for the various functions.

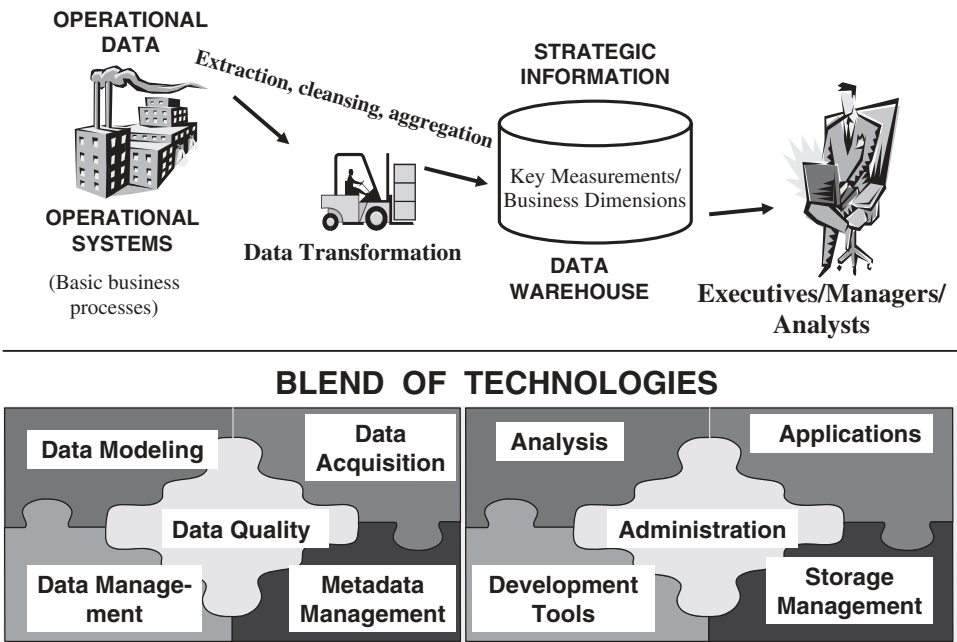


Figure 1-9 The data warehouse: a blend of technologies.

Although many technologies are in use, they all work together in a data warehouse. The end result is the creation of a new computing environment for the purpose of providing the strategic information every enterprise needs desperately.

THE DATA WAREHOUSING MOVEMENT

As enterprises began to realize the effectiveness of data warehousing, more and more organizations jumped on the bandwagon and data warehousing began to spread at a rapid rate. First the large companies that were able to quickly afford the outlay of resources began to launch data warehousing projects. Medium-sized companies also entered the data warehousing arena. Soon several businesses began to reap the benefits provided by data warehousing. Much research began to be focused on this new phenomenon. Many vendors began to offer hardware and software products to support the different functions within the data warehouse.

Prior to the data warehousing concept with an architectural model for the movement of data from operational systems to decision support environments, companies attempted multiple decision-support environments within their organizations. This had to be done with enormous costs fraught with large amounts of data redundancies and inconsistencies. Each decision-support environment was intended to serve specific groups of users for limited purposes. The adoption of data warehousing changed all of this. Similar to industrial warehouses, data warehouses are intended for large-scale collection and storage of corporate data to provide strategic information for the overall needs. Just as products stored in industrial warehouses are distributed to retail stores or marts, data stored in data warehouses may be channeled to data marts for specific users.

Data Warehousing Milestones

As data warehousing gained acceptance during the 1980s and 1990s, we may trace some of the highlights of the movement. Listed below are the significant milestones during the initial phase of the movement:

- 1983—Teradata introduces a database management system (DBMS) designed for decision-support systems.
- 1988—The article *An Architecture for a Business and Information Systems* introducing the term “business data warehouse” is published by Barry Devlin and Paul Murphy in the *IBM Systems Journal*.
- 1990—Red Brick Systems introduces Red Brick Warehouse, a DBMS specifically for data warehousing.
- 1991—Bill Inmon publishes his book *Building the Data Warehouse* (he is generally considered the father of data warehousing).
- 1991—Prism Solutions introduces Prism Warehouse Manager software for developing a data warehouse.
- 1995—The Data Warehousing Institute, a premier institution that promotes data warehousing is founded. (This institution has since emerged as the leading voice in the data warehousing and business intelligence arena providing education, research, and support.)

- 1996—Ralph Kimball publishes a seminal book *The Data Warehousing Toolkit*. (He is among the top authorities in the field of data warehousing and decision support systems.)
- 1997—Oracle 8, with support for STAR schema queries, is released.

Initial Challenges

As the adoption of data warehousing by organizations continued, even those companies that implemented data warehouses faced significant challenges that promoted a moderate shift from the original implementations. Here is a list of the key challenges that had to be overcome:

- Customers had become more sophisticated and savvy, pressing for greater service, improved quality, and innovative customization.
- Government deregulation of industries exposed companies to fiercer competition and the need for leaner operation.
- Expansion of globalization opened the arena for competitors, more in number and greater in power.
- New privacy regulations created the need to revise methods of collection and use of information.
- Improper architecture of some initial data warehousing systems produced fragmented views of corporate data and tended to produce disparate information silos.
- Query, reporting, and analysis tools provided to the users in the early data warehousing environments for self-service proved to be too complex and overwhelming for use by the users themselves.
- The promises of early data warehouse environments to provide user-friendly tools for the masses remained unfulfilled.

EVOLUTION OF BUSINESS INTELLIGENCE

The initial challenges following the adoption of early data warehousing systems forced companies to take a second look at providing decision support. Companies began to perceive that the goal of decision-support systems is twofold: transformation of data to information; derivation of knowledge from information. Each of these two aspects needs to be emphasized and strengthened appropriately to provide the necessary results. Business intelligence for an organization requires two environments, one to concentrate on transformation of data into information and the other to deal with transformation of information into knowledge.

Business intelligence (BI), therefore, is a broad group of applications and technologies. First, the term refers to the systems and technologies for gathering, cleansing, consolidating, and storing corporate data. Next, business intelligence relates to the tools, techniques, and applications for analyzing the stored data. The Gartner Group popularized BI as an umbrella term to include concepts and methods to improve business decision making by fact-based support systems. The Data Warehousing Institute compares BI to a data refinery. Similar to an oil refinery, a BI setting takes data as the raw material, collects it, refines it, and processes it into several information products.

BI: Two Environments

When you consider all that BI encompasses, you may view BI for an enterprise as composed of two environments:

Data to Information. In this environment data from multiple operational systems are extracted, integrated, cleansed, transformed and stored as information in specially designed repositories.

Information to Knowledge. In this environment analytical tools are made available to users to access and analyze the information content in the specially designed repositories and turn information into knowledge.

BI: Data Warehousing and Analytics

As some of the early challenges indicated, sufficient and separate attention needs to be given to the two environments that BI encompasses. In today's businesses, extraction, consolidation, transformation, and storing of data as strategic information is a formidable task. Again, using this information with sophisticated tools for proper decision making is equally challenging. Therefore, the trend is to consider these as two distinct environments for corporate BI. Vendors also tend to specialize in tools appropriate for these two distinct environments.

However, the two environments are complementary and need to work together. Figure 1-10 shows the two complementary environments, the data warehousing environment, which transforms data into information, and the analytical environment, which produces knowledge from information. As we proceed from chapter to chapter, we will keep expanding and intensifying our discussion of these two environments.

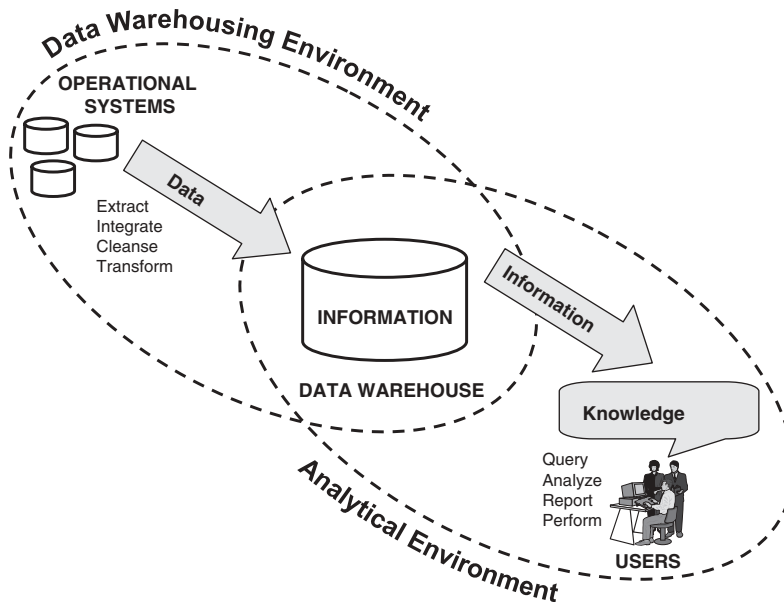


Figure 1-10 BI: data warehousing and analytical environments.

CHAPTER SUMMARY

- Companies are desperate for strategic information to counter fiercer competition, extend market share, and improve profitability.
- In spite of tons of data accumulated by enterprises over the past decades, every enterprise is caught in the middle of an information crisis. Information needed for strategic decision making is not readily available.
- All the past attempts by IT to provide strategic information have been failures. This was mainly because IT has been trying to provide strategic information from operational systems.
- Informational systems are different from the traditional operational systems. Operational systems are not designed for strategic information.
- We need a new type of computing environment to provide strategic information. The data warehouse promises to be this new computing environment.
- Data warehousing is the viable solution. There is a compelling need for data warehousing in every enterprise.
- The challenges faced in early data warehouse implementations led the movement towards maturity.
- The notion of business intelligence for an enterprise has evolved as an umbrella concept embracing data warehousing and analytics to transform data into information and information into knowledge.

REVIEW QUESTIONS

1. What do we mean by strategic information? For a commercial bank, name five types of strategic objectives.
2. Do you agree that a typical retail store collects huge volumes of data through its operational systems? Name three types of transaction data likely to be collected by a retail store in large volumes during its daily operations.
3. Examine the opportunities that can be provided by strategic information for a medical center. Can you list five such opportunities?
4. Why were all the past attempts by IT to provide strategic information failures? List three concrete reasons and explain.
5. Describe five differences between operational systems and informational systems.
6. Why are operational systems not suitable for providing strategic information? Give three specific reasons and explain.
7. Name six characteristics of the computing environment needed to provide strategic information.
8. What types of processing take place in a data warehouse? Describe.
9. A data warehouse is an environment, not a product. Discuss.

10. Data warehousing is the only viable means to resolve the information crisis and to provide strategic information. List four reasons to support this assertion and explain them.

EXERCISES

1. Match the columns:

- | | |
|----------------------------|---------------------------------|
| 1. information crisis | A. OLTP applications |
| 2. strategic information | B. produce ad hoc reports |
| 3. operational systems | C. explosive growth |
| 4. information center | D. despite lots of data |
| 5. data warehouse | E. data cleaned and transformed |
| 6. order processing | F. users go to get information |
| 7. executive information | G. used for decision making |
| 8. data staging area | H. environment, not product |
| 9. extract programs | I. for day-to-day operations |
| 10. information technology | J. simple, easy to use |

2. The current trends in hardware/software technology make data warehousing feasible. Explain with some examples how exactly technology trends do help.
3. You are the IT director of a nationwide insurance company. Write a memo to the executive vice president explaining the types of opportunities that can be realized with readily available strategic information.
4. For an airline company, how can strategic information increase the number of frequent flyers? Discuss giving specific details.
5. You are a senior analyst in the IT department of a company manufacturing automobile parts. The marketing VP is complaining about the poor response by IT in providing strategic information. Draft a proposal to him introducing the concept of business intelligence and how data warehousing and analytics as part of business intelligence for your company would be the optimal solution.

CHAPTER 2

DATA WAREHOUSE: THE BUILDING BLOCKS

CHAPTER OBJECTIVES

- Review formal definitions of a data warehouse
- Discuss the defining features
- Distinguish between data warehouses and data marts
- Review the evolved architectural types
- Study each component or building block that makes up a data warehouse
- Introduce metadata and highlight its significance

As we have seen in the last chapter, data warehousing has evolved as part of business intelligence for the enterprise. In the data warehouse you integrate and transform enterprise data into information suitable for strategic decision making. You take all the historic data from the various operational systems, combine this internal data with any relevant data from outside sources, and pull them together. You resolve any conflicts in the way data resides in different systems and transform the integrated data content into a format suitable for providing information to the various classes of users. Finally, you supplement with information delivery methods.

In order to set up this information delivery system, you need different components or building blocks. These building blocks are arranged together in the most optimal way to serve the intended purpose. They are arranged in a suitable architecture. Before we get into the individual components and their arrangement in the overall architecture, let us first look at some fundamental features of the data warehouse.

Bill Inmon (1996, p. 33), considered to be the father of data warehousing as noted in the previous chapter, provides the following definition: “A Data Warehouse is a subject oriented, integrated, nonvolatile, and time variant collection of data in support of management’s decisions.”

Sean Kelly, another leading data warehousing practitioner, defines the data warehouse in the following way. The data in the data warehouse is:

- Separate
- Available
- Integrated
- Time stamped
- Subject oriented
- Nonvolatile
- Accessible

DEFINING FEATURES

Let us examine some of the key defining features of the data warehouse based on these definitions. What about the nature of the data in the data warehouse? How is this data different from the data in any operational system? Why does it have to be different? How is the data content in the data warehouse used?

Subject-Oriented Data

In operational systems, we store data by individual applications. In the data sets for an order processing application, we keep the data for that particular application. These data sets provide the data for all the functions for entering orders, checking stock, verifying customer's credit, and assigning the order for shipment. But these data sets contain only the data that is needed for those functions relating to this particular application. We will have some data sets containing data about individual orders, customers, stock status, and detailed transactions, but all of these are structured around the processing of orders.

Similarly, for a banking institution, data sets for a consumer loans application contain data for that particular application. Data sets for other distinct applications for checking accounts and savings accounts relate to those specific applications. Again, in an insurance company, different data sets support individual applications such as automobile insurance, life insurance, and workers' compensation insurance.

In every industry, data sets are organized around individual applications to support those particular operational systems. These individual data sets have to provide data for the specific applications to perform the specific functions efficiently. Therefore, the data sets for each application need to be organized around that specific application.

In striking contrast, in the data warehouse, data is stored by real-world business subjects or events, not by applications. The data in a data warehouse is organized in such a way that all the data sets relating to the same real-world business subject or event is tied together. We have said that data is linked and stored by business subjects. Well, what are business subjects? Business subjects differ from enterprise to enterprise. These are the subjects critical for the enterprise. For a manufacturing company, sales, shipments, and inventory are critical business subjects. For a retail store, sales at the check-out counter would be a critical business subject.

Figure 2-1 distinguishes between how data is stored in operational systems and in the data warehouse. In the operational systems shown, data for each application is organized separately by application: order processing, consumer loans, customer billing, accounts

In the data warehouse, data is not stored by operational applications, but by business subjects.

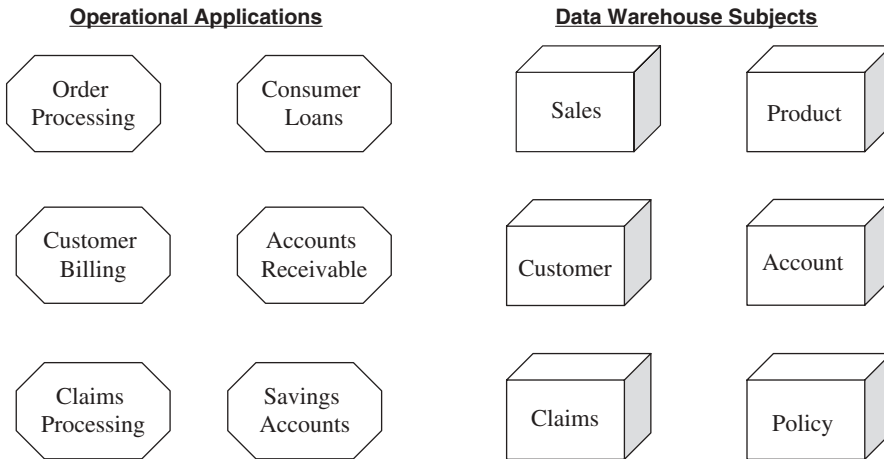


Figure 2-1 The data warehouse is subject oriented.

receivable, claims processing, and savings accounts. For example, *claims* is a critical business subject for an insurance company. Claims under automobile insurance policies are processed in the auto insurance application. Claims data for automobile insurance is organized in that application. Similarly, claims data for workers' compensation insurance is organized in the workers' comp insurance application. But in the data warehouse for an insurance company, data about *claims* is organized around the subject of *claims* and not by *individual* applications of auto insurance and workers' comp.

In a data warehouse, there is no application flavor. The data in a data warehouse cuts across applications.

Integrated Data

For proper decision making, you need to pull together all the relevant data from the various applications. The data in the data warehouse comes from several operational systems. Source data reside in different databases, files, and data segments. These are disparate applications, so the operational platforms and operating systems could be different. The file layouts, character code representations, and field naming conventions all could be different.

In addition to data from internal operational systems, for many enterprises, data from outside sources is likely to be very important. Companies such as Metro Mail, A. C. Nielsen, and IRI specialize in providing vital data on a regular basis. Your data warehouse may need data from such sources. This is one more variation in the mix of source data for a data warehouse.

Figure 2-2 illustrates a simple process of data integration for a banking institution. Here the data fed into the subject area of *account* in the data warehouse comes from three different operational applications. Even within just three applications, there could be several variations. Naming conventions could be different; attributes for data items could be different. The account number in the savings account application could be eight bytes long, but only six bytes in the checking account application.

Before the data from various disparate sources can be usefully stored in a data warehouse, you have to remove the inconsistencies. You have to standardize the various data elements

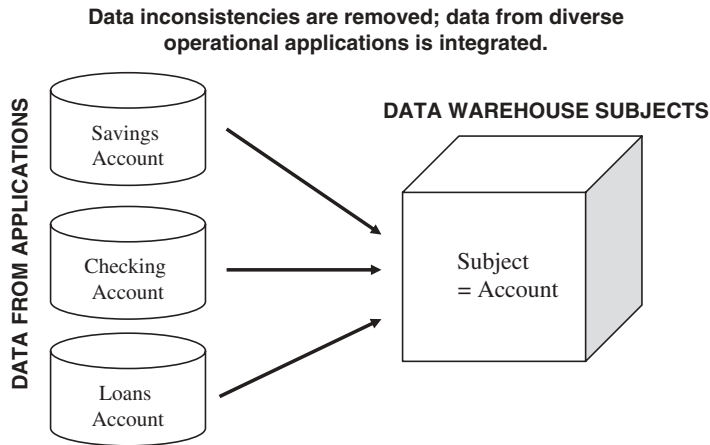


Figure 2-2 The data warehouse is integrated.

and make sure of the meanings of data names in each source application. Before moving the data into the data warehouse, you have to go through a process of transformation, consolidation, and integration of the source data.

Here are some of the items that would need to be standardized and made consistent:

- Naming conventions
- Codes
- Data attributes
- Measurements

Time-Variant Data

For an operational system, the stored data contains the *current* values. In an accounts receivable system, the balance is the current outstanding balance in the customer's account. In an order entry system, the status of an order is the current status of the order. In a consumer loans application, the balance amount owed by the customer is the current amount. Of course, we store some past transactions in operational systems, but, essentially, operational systems reflect current information because these systems support day-to-day current operations.

On the other hand, the data in the data warehouse is meant for analysis and decision making. If a user is looking at the buying pattern of a specific customer, the user needs data not only about the current purchase, but on the past purchases as well. When a user wants to find out the reason for the drop in sales in the North East division, the user needs all the sales data for that division over a period extending back in time. When an analyst in a grocery chain wants to promote two or more products together, that analyst wants sales of the selected products over a number of past quarters.

A data warehouse, because of the very nature of its purpose, has to contain historical data, not just current values. Data is stored as snapshots over past and current periods. Changes to data are tracked and recorded so that, if necessary, reports can be produced to show changes over time. Every data structure in the data warehouse contains the time element. You will find historical snapshots of the operational data in the data warehouse. This aspect of the data warehouse is quite significant for both the design and the implementation phases.

For example, in a data warehouse containing units of sale, the quantity stored in each file record or table row relates to a specific time element. Depending on the level of the details in the data warehouse, the sales quantity in a record may relate to a specific date, week, month, or quarter.

The time-variant nature of the data in a data warehouse

- Allows for analysis of the past
- Relates information to the present
- Enables forecasts for the future

Nonvolatile Data

Data extracted from the various operational systems and pertinent data obtained from outside sources are transformed, integrated, and stored in the data warehouse. The data in the data warehouse is not intended for running of the day-to-day business. When you want to process the next order received from a customer, you do not look into the data warehouse to find the current stock status. The operational order entry application is meant for that purpose. In the data warehouse, you keep the extracted stock status data as snapshots over time. You do not update the data warehouse every time you process a single order.

Data from the operational systems are moved into the data warehouse at specific intervals. Depending on the requirements of the business, these data movements take place twice a day, once a day, once a week, or maybe once in two weeks. In fact, in a typical data warehouse, data movements to different data sets may take place at different frequencies. The changes to the attributes of the products may be moved once a week. Any revisions to geographical setup may be moved once a month. The units of sales may be moved once a day. You plan and schedule the data movements or data loads based on the requirements of your users.

As illustrated in Fig. 2-3, every business transaction does not update the data in the data warehouse. The business transactions update the operational system databases in real time. We add, change, or delete data from an operational system as each transaction happens but do not usually update the data in the data warehouse. You do not delete the data in the data

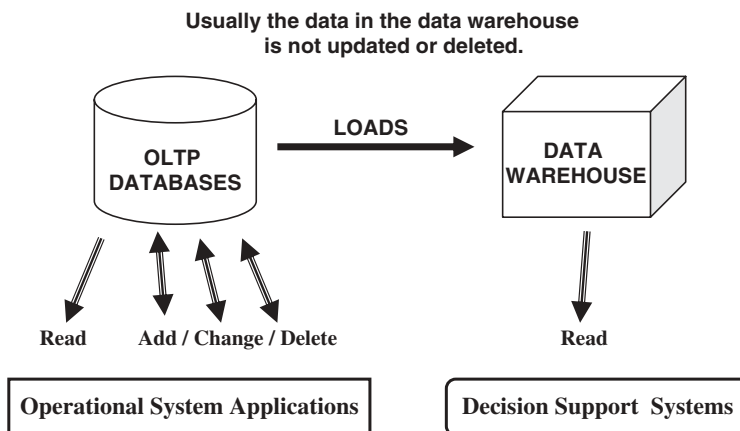


Figure 2-3 The data warehouse is nonvolatile.

warehouse in real time. Once the data is captured and committed in the data warehouse, you do not run individual transactions to change the data there. Data updates are commonplace in an operational database; not so in a data warehouse. The data in a data warehouse is not as volatile as the data in an operational database is. The data in a data warehouse is primarily for query and analysis.

Data Granularity

In an operational system, data is usually kept at the lowest level of detail. In a point-of-sale system for a grocery store, the units of sale are captured and stored at the level of units of a product per transaction at the check-out counter. In an order entry system, the quantity ordered is captured and stored at the level of units of a product per order received from the customer. Whenever you need summary data, you add up the individual transactions. If you are looking for units of a product ordered this month, you read all the orders entered for the entire month for that product and add up. You do not usually keep summary data in an operational system.

When a user queries the data warehouse for analysis, he or she usually starts by looking at summary data. The user may start with total sale units of a product in an entire region. Then the user may want to look at the breakdown by states in the region. The next step may be the examination of sale units by the next level of individual stores. Frequently, the analysis begins at a high level and moves down to lower levels of detail.

In a data warehouse, therefore, you find it efficient to keep data summarized at different levels. Depending on the query, you can then go to the particular level of detail and satisfy the query. Data granularity in a data warehouse refers to the level of detail. The lower the level of detail, the finer is the data granularity. Of course, if you want to keep data in the lowest level of detail, you have to store a lot of data in the data warehouse. You will have to decide on the granularity levels based on the data types and the expected system performance for queries. Figure 2-4 shows examples of data granularity in a typical data warehouse.

THREE DATA LEVELS IN A BANKING DATA WAREHOUSE

<u>Daily Detail</u>	<u>Monthly Summary</u>	<u>Quarterly Summary</u>
Account	Account	Account
Activity Date	Month	Quarter
Amount	Number of transactions	Number of transactions
Deposit/Withdrawal	Withdrawals	Withdrawals
	Deposits	Deposits
	Beginning Balance	Beginning Balance
	Ending Balance	Ending Balance

Data granularity refers to the level of detail. Depending on the requirements, multiple levels of detail may be present. Many data warehouses have at least dual levels of granularity.

Figure 2-4 Data granularity.

DATA WAREHOUSES AND DATA MARTS

Those who had been following the literature on data warehouses in the early years of the subject no doubt came across the terms “data warehouse” and “data mart.” Many who were new to this paradigm were confused about these terms. Some authors and vendors used the two terms synonymously. Some made distinctions that were not clear enough. At this point, it would be worthwhile for us to examine these two terms and take our position.

Writing in a leading trade magazine in 1998, the early days of data warehousing, Bill Inmon stated, “The single most important issue facing the IT manager this year is whether to build the data warehouse first or the data mart first.” This statement is somewhat true even today. Let us examine this statement and tackle the definitions.

Before deciding to build a data warehouse for your organization, you need to ask the following basic and fundamental questions and address the relevant issues:

- Top-down or bottom-up approach?
- Enterprise-wide or departmental?
- Which first—data warehouse or data mart?
- Build pilot or go with a full-fledged implementation?
- Dependent or independent data marts?

These are critical issues requiring careful examination and planning.

Should you look at the big picture of your organization, take a top-down approach, and build a mammoth data warehouse? Or, should you adopt a bottom-up approach, look at the individual local and departmental requirements, and build bite-size departmental data marts?

Should you build a large data warehouse and then let that repository feed data into local, departmental data marts? On the other hand, should you build individual local data marts, and combine them to form your overall data warehouse? Should these local data marts be independent of one another? Or should they be dependent on the overall data warehouse for data feed? Should you build a pilot data mart? These are crucial questions.

How Are They Different?

Let us take a close look at Fig. 2-5. Here are the two different basic approaches: (1) overall data warehouse feeding dependent data marts, and (2) several departmental or local data marts combining into a data warehouse. In the first approach, you extract data from the operational systems; you then transform, clean, integrate, and keep the data in the data warehouse. So, which approach is best in your case, the top-down or the bottom-up approach? Let us examine these two approaches carefully.

Top-Down Versus Bottom-Up Approach

Top-Down Approach Bill Inmon is one of the leading proponents of the top-down approach. He has defined a data warehouse as a centralized repository for the entire enterprise. In this approach the data in the data warehouse is stored at the lowest level of granularity based on a normalized data model. In the Inmon vision the data warehouse is at the center of the “Corporate Information Factory” (CIF) providing the logical framework for delivering business intelligence to the enterprise. Business operations provide data to

DATA WAREHOUSE	DATA MART
◆ Corporate/Enterprise-wide ◆ Union of all data marts ◆ Data received from staging area ◆ Queries on presentation resource ◆ Structure for corporate view of data ◆ Organized on E-R model	◆ Departmental ◆ A single business process ◆ STARjoin (facts & dimensions) ◆ Technology optimal for data access and analysis ◆ Structure to suit the departmental view of data

Figure 2-5 Data warehouse versus data mart.

drive the CIF. The centralized data warehouse would feed the dependent data marts that may be designed based on a dimensional data model.

The advantages of this approach are:

- A truly corporate effort, an enterprise view of data
- Inherently architected, not a union of disparate data marts
- Single, central storage of data about the content
- Centralized rules and control
- May see quick results if implemented with iterations

The disadvantages are:

- Takes longer to build even with an iterative method
- High exposure to risk of failure
- Needs high level of cross-functional skills
- High outlay without proof of concept

This is the big-picture approach in which you build the overall, big, enterprise-wide data warehouse. Here you do not have a collection of fragmented islands of information. The data warehouse is large and integrated. This approach, however, would take longer to build and has a high risk of failure. If you do not have experienced professionals on your team, this approach could be hazardous. Also, it will be difficult to sell this approach to senior management and sponsors. They are not likely to see results soon enough.

Bottom-Up Approach Ralph Kimball, another leading author and expert practitioner in data warehousing, is a proponent of the approach that has come to be known as the bottom-up approach. Kimball (1996) envisions the corporate data warehouse as a collection of conformed data marts. The key consideration is the conforming of the dimensions among the separate data marts. In this approach data marts are created first to provide analytical and reporting capabilities for specific business subjects based on the dimensional data model.

Data marts contain data at the lowest level of granularity and also as summaries depending on the needs for analysis. These data marts are joined or “unioned” together by conforming the dimensions. We will discuss this process of conforming the dimensions in great detail in subsequent chapters.

The advantages of this approach are:

- Faster and easier implementation of manageable pieces
- Favorable return on investment and proof of concept
- Less risk of failure
- Inherently incremental; can schedule important data marts first
- Allows project team to learn and grow

The disadvantages are:

- Each data mart has its own narrow view of data
- Permeates redundant data in every data mart
- Perpetuates inconsistent and irreconcilable data
- Proliferates unmanageable interfaces

In this bottom-up approach, you build your departmental data marts one by one. You would set a priority scheme to determine which data marts you must build first. The most severe drawback of this approach is data fragmentation. Each independent data mart will be blind to the overall requirements of the entire organization.

A Practical Approach

In order to formulate an approach for your organization, you need to examine what exactly your organization wants. Is your organization looking for long-term results or fast data marts for only a few subjects for now? Does your organization want quick, proof-of-concept, throw-away implementations? Or, do you want to look into some other practical approach?

Although the top-down and the bottom-up approaches each have their own advantages and drawbacks, a compromise approach accommodating both views appears to be practical. In this approach we do not lose sight of the overall big picture for the entire enterprise. We base our planning on this overall big picture. This aspect is from the top-down approach. Then we adopt the principles of the bottom-up approach and build the conformed data marts based on a priority scheme. The steps in this practical approach are as follows:

1. Plan and define requirements at the overall corporate level
2. Create a surrounding architecture for a complete warehouse
3. Conform and standardize the data content
4. Implement the data warehouse as a series of supermarts, one at a time

In this practical approach, you go to the basics and determine what exactly your organization wants in the long term. The key to this approach is that you first plan at the enterprise level. You gather requirements at the overall level. You establish the architecture for the complete warehouse. Then you determine the data content for each supermart. Supermarts

are carefully architected data marts. You implement these supermarts, one at a time. Before implementation, you make sure that the data content among the various supermarts are conformed in terms of data types, field lengths, precision, and semantics. A certain data element must mean the same thing in every supermart. This will avoid spread of disparate data across several data marts.

A data mart, in this practical approach, is a logical subset of the complete data warehouse, a sort of pie-wedge of the whole data warehouse. A data warehouse, therefore, is a conformed union of all data marts. Individual data marts are targeted to particular business groups in the enterprise, but the collection of all the data marts form an integrated whole, called the enterprise data warehouse.

When we refer to data warehouses and data marts in our discussions here, we use the meanings as understood in this practical approach. For us, a data warehouse means a collection of the constituent data marts.

ARCHITECTURAL TYPES

We have now reviewed the basic definitions and features of data warehouses and data marts and completed a significant discussion of them. We have established our position on what the terms data warehouse and data mart mean to us. Now we are almost ready to examine the individual components.

Over the years, in practice, several arrangements of data warehouses and data marts have evolved based on the individual requirements of organizations. Before we proceed to study the individual components within a data warehouse, let us briefly discuss the leading architectural types and note how individual data marts feature in these architectural types. For each of these architectural types we will indicate how data is stored in the data warehouse and the relationships between the data warehouse and the data marts. Figure 2-6 illustrates each of these architectural types.

Centralized Data Warehouse

This architectural type takes into account the enterprise-level information requirements. An overall infrastructure is established. Atomic level normalized data at the lowest level of granularity is stored in the third normal form. Occasionally, some summarized data is included. Queries and applications access the normalized data in the central data warehouse. There are no separate data marts.

Independent Data Marts

This architectural type evolves in companies where the organizational units develop their own data marts for their own specific purposes. Although each data mart serves the particular organizational unit, these separate data marts do not provide “a single version of the truth.” The data marts are independent of one another. As a result, these different data marts are likely to have inconsistent data definitions and standards. Such variances hinder analysis of data across data marts. For example, if there are two independent data marts, one for sales and the other for shipments, although sales and shipments are related subjects, the independent data marts would make it difficult to analyze sales and shipments data together.

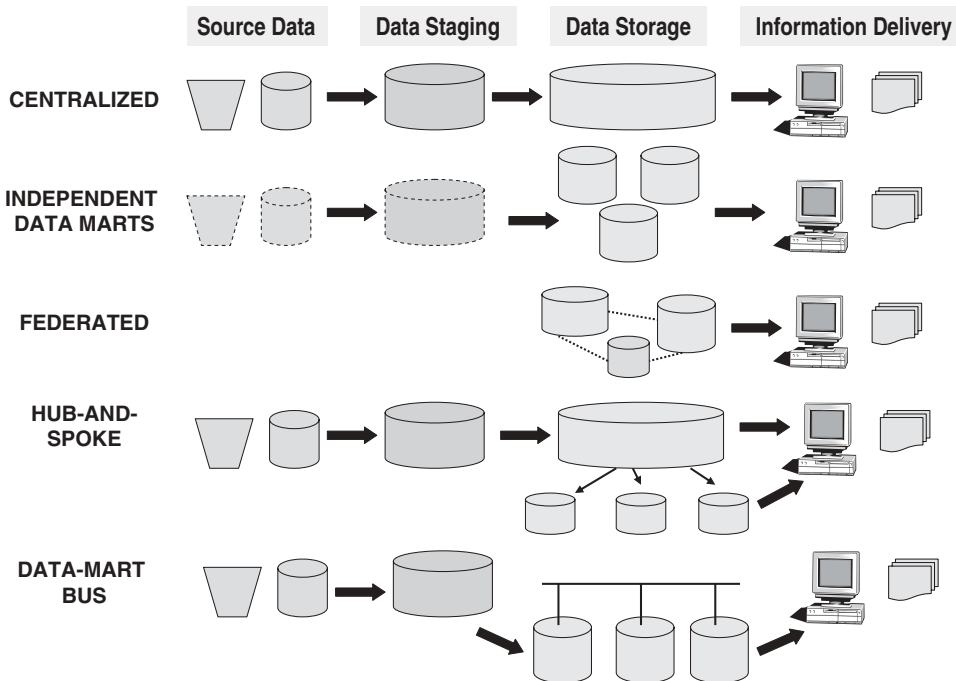


Figure 2-6 Data warehouse architectural types.

Federated

Some companies get into data warehousing with an existing legacy of an assortment of decision-support structures in the form of operational systems, extracted datasets, primitive data marts, and so on. For such companies, it may not be prudent to discard all that huge investment and start from scratch. The practical solution is a federated architectural type where data may be physically or logically integrated through shared key fields, overall global metadata, distributed queries, and such other methods. In this architectural type, there is no one overall data warehouse.

Hub-and-Spoke

This is the Inmon Corporate Information Factory approach. Similar to the centralized data warehouse architecture, here too is an overall enterprise-wide data warehouse. Atomic data in the third normal form is stored in the centralized data warehouse. The major and useful difference is the presence of dependent data marts in this architectural type. Dependent data marts obtain data from the centralized data warehouse. The centralized data warehouse forms the hub to feed data to the data marts on the spokes. The dependent data marts may be developed for a variety of purposes: departmental analytical needs, specialized queries, data mining, and so on. Each dependent data mart may have normalized, denormalized, summarized, or dimensional data structures based on individual requirements. Most queries are directed to the dependent data marts although the centralized data warehouse may itself be used for querying. This architectural type results from adopting a top-down approach to data warehouse development.

Data-Mart Bus

This is the Kimbal conformed supermarts approach. You begin with analyzing requirements for a specific business subject such as orders, shipments, billings, insurance claims, car rentals, and so on. You build the first data mart (supermart) using business dimensions and metrics. These business dimensions will be shared in the future data marts. The principal notion is that by conforming dimensions among the various data marts, the result would be logically integrated supermarts that will provide an enterprise view of the data. The data marts contain atomic data organized as a dimensional data model. This architectural type results from adopting an enhanced bottom-up approach to data warehouse development.

OVERVIEW OF THE COMPONENTS

Having reviewed a few typical ways of organizing data warehouses and data marts, we are now ready to launch a general discussion of the components of the data warehouse.

When we build an operational system such as order entry, claims processing, or savings account, we put together several components to make up the system. The front-end component consists of the GUI (graphical user interface) to interface with the users for data input. The data storage component includes the database management system, such as Oracle, Informix, or Microsoft SQL Server. The display component is the set of screens and reports for the users. The data interfaces and the network software form the connectivity component. Depending on the information requirements and the framework of our organization, we arrange these components in the most optimum way.

Architecture is the proper arrangement of the components. You build a data warehouse with software and hardware components. To suit the requirements of your organization you arrange these building blocks in a certain way for maximum benefit. You may want to lay special emphasis on one component; you may want to bolster up another component with extra tools and services. All of this depends on your circumstances.

Figure 2-7 shows the basic components of a typical warehouse. You see the *Source Data* component shown on the left. The *Data Staging* component serves as the next building block. In the middle, you see the *Data Storage* component that manages the data warehouse data. This component not only stores and manages the data, it also keeps track of the data by means of the metadata repository. The *Information Delivery* component shown on the right consists of all the different ways of making the information from the data warehouse available to the users.

Whether you build a data warehouse for a large manufacturing company on the Fortune 500 list, a leading grocery chain with stores all over the country, or a global banking institution, the basic components are the same. Each data warehouse is put together with the same building blocks. The essential difference for each organization is in the way these building blocks are arranged. The variation is in the manner in which some of the blocks are made stronger than others in the architecture.

We will now take a closer look at each of the components. At this stage, we want to know what the components are and how each fits into the architecture. We also want to review specific issues relating to each particular component.

Source Data Component

Source data coming into the data warehouse may be grouped into four broad categories, as discussed here.

Architecture is the proper arrangement of the components.

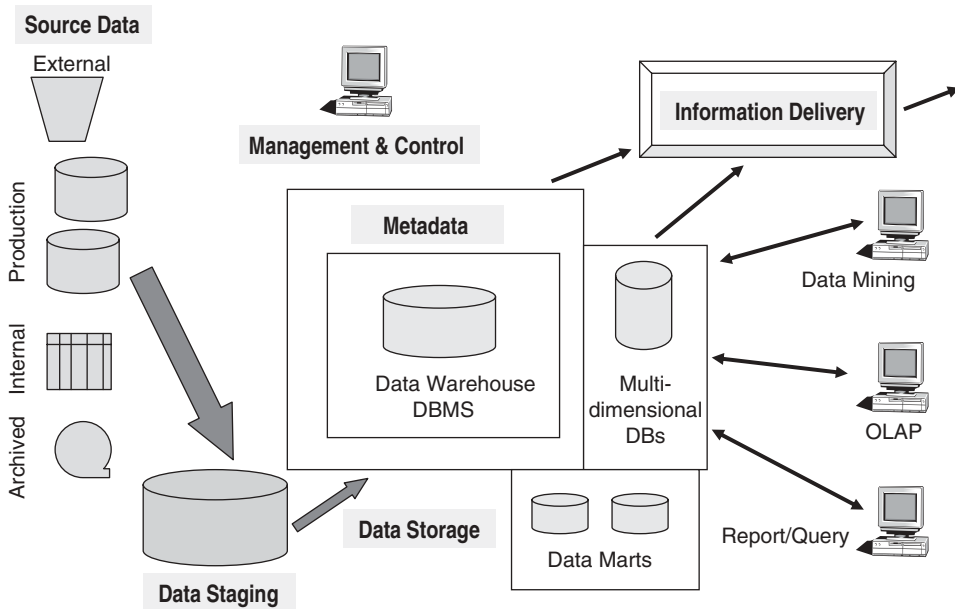


Figure 2-7 Data warehouse: building blocks or components.

Production Data This category of data comes from the various operational systems of the enterprise. These normally include financial systems, manufacturing systems, systems along the supply chain, and customer relationship management systems. Based on the information requirements in the data warehouse, you choose segments of data from the different operational systems. While dealing with this data, you come across many variations in the data formats. You also notice that the data resides on different hardware platforms. Further, the data is supported by different database systems and operating systems. This is data from many vertical applications.

In operational systems, information queries are narrow. You query an operational system for information about specific instances of business objects. You may want just the name and address of a single customer. Or, you may need the orders placed by a single customer in a single week. Or, you may just need to look at a single invoice and the items billed on that single invoice. In operational systems, you do not have broad queries. You do not query the operational system in unexpected ways. The queries are all predictable. Again, you do not expect a particular query to run across different operational systems. What does all of this mean? Simply this: there is no conformance of data among the various operational systems of an enterprise. A term like *an account* may have different meanings in different systems.

The significant and disturbing characteristic of production data is disparity. Your great challenge is to standardize and transform the disparate data from the various production systems, convert the data, and integrate the pieces into useful data for storage in the data warehouse. It is really the integration of these various sources that provide the value to the data in the data warehouse.

Internal Data In every organization, users keep their “private” spreadsheets, documents, customer profiles, and sometimes even departmental databases. This is the internal data, parts of which could be useful in a data warehouse.

If your organization does business with customers on a one-to-one basis and the contribution of each customer to the bottom line is significant, then detailed customer profiles with ample demographics are important in a data warehouse. Profiles of individual customers become very important for consideration. When your account representatives talk to their assigned customers or when your marketing department wants to make specific offerings to individual customers, you need the details. Although much of this data may be extracted from production systems, a lot of it is held by individuals and departments in their private files.

You cannot ignore the internal data held in private files in your organization. It is a collective judgment call on how much of the internal data should be included in the data warehouse. The IT department must work with the user departments to gather the internal data.

Internal data adds additional complexity to the process of transforming and integrating the data before it can be stored in the data warehouse. You have to determine strategies for collecting data from spreadsheets, find ways of taking data from textual documents, and tie into departmental databases to gather pertinent data from those sources. Again, you may want to schedule the acquisition of internal data. Initially, you may want to limit yourself to only some significant portions before going live with your first data mart.

Archived Data Operational systems are primarily intended to run the current business. In every operational system, you periodically take the old data and store it in archived files. The circumstances in your organization dictate how often and which portions of the operational databases are archived for storage. Some data is archived after a year. Sometimes data is left in the operational system databases for as long as five years. Much of the archived data comes from old legacy systems that are nearing the end of their useful lives in organizations.

Many different methods of archiving exist. There are staged archival methods. At the first stage, recent data is archived to a separate archival database that may still be online. At the second stage, the older data is archived to flat files on disk storage. At the next stage, the oldest data is archived to tape cartridges or microfilm and even kept off-site.

As mentioned earlier, a data warehouse keeps historical snapshots of data. You essentially need historical data for analysis over time. For getting historical information, you look into your archived data sets. Depending on your data warehouse requirements, you have to include sufficient historical data. This type of data is useful for discerning patterns and analyzing trends.

External Data Most executives depend on data from external sources for a high percentage of the information they use. They use statistics relating to their industry produced by external agencies and national statistical offices. They use market share data of competitors. They use standard values of financial indicators for their business to check on their performance.

For example, the data warehouse of a car rental company contains data on the current production schedules of the leading automobile manufacturers. This external data in the data warehouse helps the car rental company plan for its fleet management.

The purposes served by such external data sources cannot be fulfilled by the data available within your organization itself. The insights gleaned from your production data and your archived data are somewhat limited. They give you a picture based on what you are doing or have done in the past. In order to spot industry trends and compare performance against other organizations, you need data from external sources.

Usually, data from outside sources do not conform to your formats. You have to devise ways to convert data into your internal formats and data types. You have to organize the data transmissions from the external sources. Some sources may provide information at regular, stipulated intervals. Others may give you the data on request. You need to accommodate the variations.

Data Staging Component

After you have extracted data from various operational systems and from external sources, you have to prepare the data for storing in the data warehouse. The extracted data coming from several disparate sources needs to be changed, converted, and made ready in a format that is suitable to be stored for querying and analysis.

Three major functions need to be performed for getting the data ready. You have to extract the data, transform the data, and then load the data into the data warehouse storage. These three major functions of extraction, transformation, and preparation for loading take place in a staging area. The data staging component consists of a workbench for these functions. Data staging provides a place and an area with a set of functions to clean, change, combine, convert, deduplicate, and prepare source data for storage and use in the data warehouse.

Why do you need a separate place or component to perform the data preparation? Can you not move the data from the various sources into the data warehouse storage itself and then prepare the data? When we implement an operational system, we are likely to pick up data from different sources, move the data into the new operational system database, and run data conversions. Why can't this method work for a data warehouse? The essential difference here is this: in a data warehouse you pull in data from many source operational systems. Remember that data in a data warehouse is subject-oriented and cuts across operational applications. A separate staging area, therefore, is a necessity for preparing data for the data warehouse. Occasionally, particularly with old legacy systems, you may need a separate literal staging area. What you have in this literal staging area would be literal copies of source system content but in a more convenient environment (such as an ODBC accessible relational database). This literal staging area then acts as a surrogate for the particular source systems.

Now that we have clarified the need for a separate data staging component, let us understand what happens in data staging. We will now briefly discuss the three major functions that take place in the staging area.

Data Extraction This function has to deal with numerous data sources. You have to employ the appropriate technique for each data source. Source data may be from different source machines in diverse data formats. Part of the source data may be in relational database systems. Some data may be on other legacy network and hierarchical data models. Many data sources may still be in flat files. You may want to include data from spreadsheets and local departmental data sets. Data extraction may become quite complex.

Tools are available on the market for data extraction. You may want to consider using outside tools suitable for certain data sources. For the other data sources, you may want to develop in-house programs to do the data extraction. Purchasing outside tools may entail high initial costs. In-house programs, on the other hand, may mean ongoing costs for development and maintenance.

After you extract the data, where do you keep the data for further preparation? You may perform the extraction function in the legacy platform itself if that approach suits your

framework. More frequently, data warehouse implementation teams extract the source into a separate physical environment from which moving the data into the data warehouse would be easier. In the separate environment, you may extract the source data into a group of flat files, or a data-staging relational database, or a combination of both.

Data Transformation In every system implementation, data conversion is an important function. For example, when you implement an operational system such as a magazine subscription application, you have to initially populate your database with data from the prior system records. You may be converting over from a manual system. Or, you may be moving from a file-oriented system to a modern system supported with relational database tables. In either case, you will convert the data from the prior systems. So, what is so different for a data warehouse? How is data transformation for a data warehouse more involved than for an operational system?

Again, as you know, data for a data warehouse comes from many disparate sources. If data extraction for a data warehouse poses great challenges, data transformation presents even greater challenges. Another factor in the data warehouse is that the data feed is not just an initial load. You will have to continue to pick up the ongoing changes from the source systems. Any transformation tasks you set up for the initial load will be adapted for the ongoing revisions as well.

You perform a number of individual tasks as part of data transformation. First, you clean the data extracted from each source. Cleaning may just be correction of misspellings, or may include resolution of conflicts between state codes and zip codes in the source data, or may deal with providing default values for missing data elements, or elimination of duplicates when you bring in the same data from multiple source systems.

Standardization of data elements forms a large part of data transformation. You standardize the data types and field lengths for same data elements retrieved from the various sources. Semantic standardization is another major task. You resolve synonyms and homonyms. When two or more terms from different source systems mean the same thing, you resolve the synonyms. When a single term means many different things in different source systems, you resolve the homonym.

Data transformation involves many forms of combining pieces of data from the different sources. You combine data from a single source record or related data elements from many source records. On the other hand, data transformation also involves purging source data that is not useful and separating out source records into new combinations. Sorting and merging of data takes place on a large scale in the data staging area.

In many cases, the keys chosen for the operational systems are field values with built-in meanings. For example, the product key value may be a combination of characters indicating the product category, the code of the warehouse where the product is stored, and some code to show the production batch. Primary keys in the data warehouse cannot have built-in meanings. We will discuss this further in Chapter 10. Data transformation also includes the assignment of surrogate keys derived from the source system primary keys.

A grocery chain point-of-sale operational system keeps the unit sales and revenue amounts by individual transactions at the check-out counter at each store. But in the data warehouse, it may not be necessary to keep the data at this detailed level. You may want to summarize the totals by product at each store for a given day and keep the summary totals of the sale units and revenue in the data warehouse storage. In such cases, the data transformation function would include appropriate summarization.

- ◆ This function is time-consuming
- ◆ Initial load moves very large volumes of data
- ◆ The business conditions determine the refresh cycles

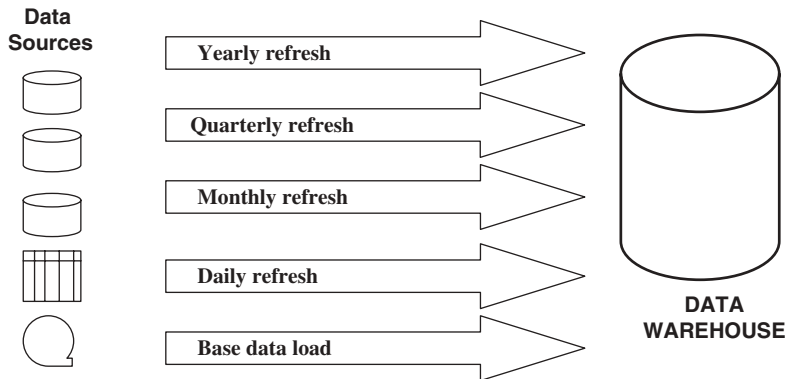


Figure 2-8 Data movements to the data warehouse.

When the data transformation function ends, you have a collection of integrated data that is cleaned, standardized, and summarized. You now have data ready to load into each data set in your data warehouse.

Data Loading Two distinct groups of tasks form the data loading function. When you complete the design and construction of the data warehouse and go live for the first time, you do the initial loading of the data into the data warehouse storage. The initial load moves large volumes of data using up substantial amounts of time. As the data warehouse starts functioning, you continue to extract the changes to the source data, transform the data revisions, and feed the incremental data revisions on an ongoing basis. Figure 2-8 illustrates the common types of data movements from the staging area to the data warehouse storage.

Data Storage Component

The data storage for the data warehouse is a separate repository. The operational systems of your enterprise support the day-to-day operations. These are online transaction processing applications. The data repositories for the operational systems typically contain only the current data. Also, these data repositories contain the data structured in highly normalized formats for fast and efficient processing. In contrast, in the data repository for a data warehouse, you need to keep large volumes of historical data for analysis. Further, you have to keep the data in the data warehouse in structures suitable for analysis, and not for quick retrieval of individual pieces of information. Therefore, the data storage for the data warehouse is kept separate from the data storage for operational systems.

In your databases supporting operational systems, the updates to data happen as transactions occur. These transactions hit the databases in a random fashion. How and when the transactions change the data in the databases is not completely within your control. The data in the operational databases could change from moment to moment. When your

analysts use the data in the data warehouse for analysis, they need to know that the data is stable and that it represents snapshots at specified periods. As they are working with the data, the data storage must not be in a state of continual updating. For this reason, the data warehouses are “read-only” data repositories.

Generally, the database in your data warehouse must be open. Depending on your requirements, you are likely to use tools from multiple vendors. The data warehouse must be open to different tools. Most of the data warehouses employ relational database management systems.

Many data warehouses also employ multidimensional database management systems. Data extracted from the data warehouse storage is aggregated in many ways and the summary data is kept in the multidimensional databases (MDDBs). Such multidimensional database systems are usually proprietary products.

Information Delivery Component

Who are the users that need information from the data warehouse? The range is fairly comprehensive. The novice user comes to the data warehouse with no training and, therefore, needs prefabricated reports and preset queries. The casual user needs information once in a while, not regularly. This type of user also needs prepackaged information. The business analyst looks for ability to do complex analysis using the information in the data warehouse. The power user wants to be able to navigate throughout the data warehouse, pick up interesting data, format his or her own queries, drill through the data layers, and create custom reports and ad hoc queries.

In order to provide information to the wide community of data warehouse users, the information delivery component includes different methods of information delivery. Figure 2-9 shows the different information delivery methods. Ad hoc reports are predefined reports primarily meant for novice and casual users. Provision for complex queries, multidimensional (MD) analysis, and statistical analysis cater to the needs of the business analysts and power users. Information fed into executive information systems (EIS) is meant for senior

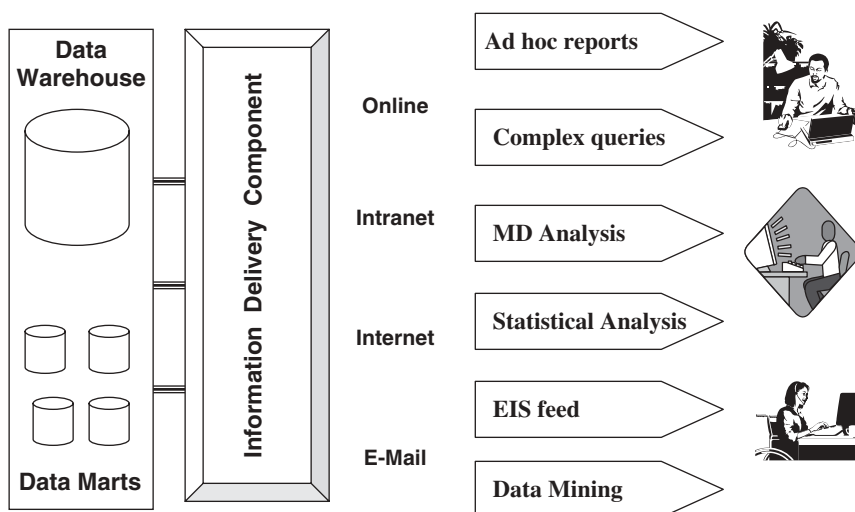


Figure 2-9 Information delivery component.

executives and high-level managers. Some data warehouses also provide data to data-mining applications. Data-mining applications are knowledge discovery systems where the mining algorithms help you discover trends and patterns from the usage of your data.

In your data warehouse, you may include several information delivery mechanisms. Most commonly, you provide for online queries and reports. The users will enter their requests online and will receive the results online. You may set up delivery of scheduled reports through e-mail or you may make adequate use of your organization's intranet for information delivery. Recently, information delivery over the Internet has been gaining ground.

Metadata Component

Metadata in a data warehouse is similar to the data dictionary or the data catalog in a database management system. In the data dictionary, you keep the information about the logical data structures, the information about the files and addresses, the information about the indexes, and so on. The data dictionary contains data about the data in the database.

Similarly, the metadata component is the data about the data in the data warehouse. This definition is a commonly used definition. We need to elaborate on this definition. Metadata in a data warehouse is similar to a data dictionary, but much more than a data dictionary. Later, in a separate section in this chapter, we will devote more time to the discussion of metadata. Here, for the sake of completeness, we just want to list metadata as one of the components of the data warehouse architecture.

Management and Control Component

This component of the data warehouse architecture sits on top of all the other components. The management and control component coordinates the services and activities within the data warehouse. This component controls the data transformation and the data transfer into the data warehouse storage. On the other hand, it moderates the information delivery to the users. It works with the database management systems and enables data to be properly stored in the repositories. It monitors the movement of data into the staging area and from there into the data warehouse storage itself.

The management and control component interacts with the metadata component to perform the management and control functions. As the metadata component contains information about the data warehouse itself, the metadata is the source of information for the management module.

METADATA IN THE DATA WAREHOUSE

Think of metadata as the Yellow Pages[®] of your town. Do you need information about the stores in your town, where they are, what their names are, and what products they specialize in? Go to the Yellow Pages. The Yellow Pages is a directory with data about the institutions in your town. Almost in the same manner, the metadata component serves as a directory of the contents of your data warehouse.

Because of the importance of metadata in a data warehouse, we have set apart all of Chapter 9 for this topic. At this stage, we just want to get an introduction to the topic and highlight that metadata is a key architectural component of the data warehouse.

Types of Metadata

Metadata in a data warehouse fall into three major categories:

- Operational metadata
- Extraction and transformation metadata
- End-user metadata

Operational Metadata As you know, data for the data warehouse comes from several operational systems of the enterprise. These source systems contain different data structures. The data elements selected for the data warehouse have various field lengths and data types. In selecting data from the source systems for the data warehouse, you split records, combine parts of records from different source files, and deal with multiple coding schemes and field lengths. When you deliver information to the end-users, you must be able to tie that back to the original source data sets. Operational metadata contain all of this information about the operational data sources.

Extraction and Transformation Metadata Extraction and transformation metadata contain data about the extraction of data from the source systems, namely, the extraction frequencies, extraction methods, and business rules for the data extraction. Also, this category of metadata contains information about all the data transformations that take place in the data staging area.

End-User Metadata The end-user metadata is the navigational map of the data warehouse. It enables the end-users to find information from the data warehouse. The end-user metadata allows the end-users to use their own business terminology and look for information in those ways in which they normally think of the business.

Special Significance

Why is metadata especially important in a data warehouse?

- First, it acts as the glue that connects all parts of the data warehouse.
- Next, it provides information about the contents and structures to the developers.
- Finally, it opens the door to the end-users and makes the contents recognizable in their own terms.

CHAPTER SUMMARY

- Defining features of the data warehouse are: separate, subject-oriented, integrated, time-variant, and nonvolatile.
- You may use a top-down approach and build a large, comprehensive, enterprise data warehouse, or you may use a bottom-up approach and build small, independent,

departmental data marts. In spite of some advantages, both approaches have serious shortcomings.

- A viable practical approach is to build conformed data marts, which together form the corporate data warehouse.
- Data warehouse building blocks or components are source data, data staging, data storage, information delivery, metadata, and management and control.
- In a data warehouse, metadata is especially significant because it acts as the glue holding all the components together and serves as a roadmap for the end-users.

REVIEW QUESTIONS

1. Name at least six characteristics or features of a data warehouse.
2. Why is data integration required in a data warehouse, more so than in an operational application?
3. Every data structure in the data warehouse contains the time element. Why?
4. Explain data granularity and how it is applicable to the data warehouse.
5. How are the top-down and bottom-up approaches for building a data warehouse different? List the major types of architectures and highlight the features of any two of these.
6. What are the various data sources for the data warehouse?
7. Why do you need a separate data staging component?
8. Under data transformation, list five different functions you can think of.
9. Name any six different methods for information delivery.
10. What are the three major types of metadata in a data warehouse? Briefly mention the purpose of each type.

EXERCISES

1. Match the columns:

- | | |
|--------------------------|-----------------------------------|
| 1. nonvolatile data | A. roadmap for users |
| 2. dual data granularity | B. subject-oriented |
| 3. dependent data mart | C. knowledge discovery |
| 4. disparate data | D. private spreadsheets |
| 5. decision support | E. application flavor |
| 6. data staging | F. because of multiple sources |
| 7. data mining | G. details and summary |
| 8. metadata | H. read-only |
| 9. operational systems | I. workbench for data integration |
| 10. internal data | J. data from main data warehouse |

2. A data warehouse is subject-oriented. What would be the major critical business subjects for the following companies?
 - a. an international manufacturing company
 - b. a local community bank
 - c. a domestic hotel chain
3. You are the data analyst on the project team building a data warehouse for an insurance company. List the possible data sources from which you will bring the data into your data warehouse. State your assumptions.
4. For an airlines company, identify three operational applications that would feed into the data warehouse. What would be the data load and refresh cycles?
5. Prepare a table showing all the potential users and information delivery methods for a data warehouse supporting a large national grocery chain.